Control Barrier Function-Based Inlet Unstart Protection for an Air-Breathing Hypersonic Vehicle

T. H. Mueller* and S. Theodoulis †
Delft University of Technology, Delft, 2628CD, Netherlands

I. Sarras‡
ONERA - The French Aerospace Lab, Palaiseau, 91123, France

J. Autenrieb§
German Aerospace Center (DLR), 38108, Braunschweig, Germany

The scramjet propulsion system of air-breathing hypersonic vehicles places stringent requirements on the vehicle states, as excessive angle-of-attack magnitudes and rates can trigger inlet unstart, an abrupt disruption of the scramjet shock system that results in immediate loss of thrust and vehicle divergence. The Generic Hypersonic Aerodynamics Model Example (GHAME) is used as a representative benchmark for this class of vehicle, and its previously developed NDI/INDI hierarchical flight control architecture is extended by integrating a Control Barrier Function (CBF) safety filter directly into the control cascade, providing a formally certified guarantee that the elevator deflection command never drives the angle-of-attack rate beyond the inlet-unstart-critical bound. The safety filter acts at the output of the angular rate loop, immediately before the actuator, ensuring the safety guarantee cannot be undone by any upstream inversion step. It is transparent to the existing architecture during normal operation, intervening minimally only when the constraint boundary is approached. The framework is implemented and evaluated in MATLAB and Simulink, and is expected to demonstrate that the inlet unstart constraint is enforced when the bound is approached, with only a minimal impact on tracking performance during active constraint intervention, and no impact during normal operation within the safe set.

Nomenclature

*MSc, student, Control & Simulation division, Faculty of Aerospace Engineering, P.O. Box 5058, 2600GB Delft, Netherlands; thomas-harald.mueller1@outlook.de.

†Associate Professor, Control & Simulation division, Faculty of Aerospace Engineering, P.O. Box 5058, 2600GB Delft, Netherlands; S.Theodoulis@tudelft.nl. Associate Fellow AIAA.

‡Research engineer, Information Processing and Systems Department, ONERA, Univ. Paris-Saclay, F-91123 Palaiseau, France; ioannis.sarras@onera.fr.

§Research Scientist, Institute of Flight Systems, Department of Flight Dynamics and Simulation; johannes.autenrieb@dlr.de (Corresponding Author).

α, μ, β	=	AoA, bank, sideslip angles	$\dot{\alpha}_{\min}, \dot{\alpha}_{\max}$	=	AoA rate bounds
$\alpha_{\min}, \alpha_{\max}$	=	AoA envelope bounds	$\dot{\alpha}_{\text{est}}$	=	Estimated AoA rate
p, q, r	=	Inertial angular rates	χ, γ	=	Heading and flight-path angle
$\delta_a, \delta_e, \delta_r$	=	Aileron, elevator, rudder defl.	δ_{vl}, δ_{vr}	=	Left/right elevon defl.
$\delta_e^{\text{cmd}}, \delta_e^{\text{safe}}$	=	Nominal and safe elevator cmd.	$\delta_{e,\min}, \delta_{e,\max}$	=	Elevator deflection limits
ρ	=	Air density	$\bar{q}$	=	Dynamic pressure
g	=	Gravitational acceleration	R_0	=	Mean Earth radius
M	=	Mach number	h	=	Altitude
V	=	Airspeed	S	=	Reference surface area
m	=	Vehicle mass	W	=	Vehicle weight
L, Y	=	Aerodynamic lift and side force	C_L, C_Y, C_D	=	Stability-axis force coeffs.
$C_{L_0}, C_{L_\alpha}, C_{L_{\delta_e}}$	=	Lift coefficients	$\omega_{\text{act}}, \zeta_{\text{act}}$	=	Act. natural freq. and damp.
$\boldsymbol{\eta}$	=	Aerodynamic angle vector	$\boldsymbol{\omega}$	=	Body rate vector
$\boldsymbol{e}_\eta, \boldsymbol{e}_\omega$	=	Attitude and rate error vectors	K_η, K_ω	=	Attitude and rate gain matrices
$T_F(\boldsymbol{x})$	=	Force contribution matrix	$T_\omega(\alpha, \beta)$	=	Body rate mapping matrix
$\boldsymbol{\nu}_{\text{cmd}}$	=	Cmd. virtual angular accel.	$\boldsymbol{\nu}_0$	=	Estimated angular accel.
$\Delta\boldsymbol{\nu}_{\text{cmd}}$	=	Incremental virtual control	$\Delta\boldsymbol{Q}_{\text{cmd}}$	=	Commanded moment increment
B	=	Control effectiveness matrix	$\boldsymbol{\delta}_{\text{cmd}}, \boldsymbol{\delta}_0$	=	Cmd. and current actuator state
$\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{y}$	=	State, input, output vectors	$\boldsymbol{f}_\nu, \boldsymbol{G}_\nu$	=	Drift vector and effectiveness matrix
$\boldsymbol{\tau}_{\text{des}}$	=	Desired virtual control input	r_i	=	Relative degree of output i
$L_f h, L_g h$	=	Lie derivatives of h	C	=	Safe set
$h(\cdot)$	=	Barrier function	ψ_i	=	HOCBF sequence functions
k_1	=	CBF gain	τ	=	CBF time constant
$\Delta\boldsymbol{u}$	=	Incremental control input	Δt	=	Sample time
I	=	Moment of inertia matrix	$\boldsymbol{\nu}$	=	Pseudo-control vector

I. Introduction

Air-breathing hypersonic vehicles operating with scramjet propulsion represent a challenging control problem in aerospace engineering. These vehicles combine highly nonlinear flight dynamics, strong aero-propulsive coupling, severe aerodynamic uncertainty, and tight propulsion-related operating constraints. In particular, the scramjet propulsion system imposes strict requirements on the admissible aerodynamic state of the vehicle, since the inlet flow must remain compatible with stable shock-system operation throughout the flight envelope [1].

A key limitation of scramjet-powered flight is the susceptibility of the inlet to unstart. Unlike conventional aircraft or ballistic re-entry vehicles, air-breathing hypersonic vehicles must maintain precise control of the incidence conditions imposed on the inlet flow. Excessive angle-of-attack magnitudes can alter the captured flow field, while rapid angle-of-attack variations can destabilise the internal shock system. If these incidence conditions exceed the admissible inlet-operating range, inlet unstart may occur, leading to a sudden disruption of the compression process, severe thrust loss, and potentially loss of vehicle control [2]. The control problem is further complicated by pronounced nonlinearities at high Mach numbers, significant uncertainty in the aerodynamic and propulsive models, and aeroelastic effects associated with slender hypersonic configurations [3]. These characteristics make robust flight control essential, but also make purely model-based safety guarantees difficult to obtain over the full operating envelope.

Nonlinear Dynamic Inversion (NDI) and Incremental Nonlinear Dynamic Inversion (INDI) provide attractive frameworks for hypersonic flight control [4]. NDI exploits the known nonlinear structure of the vehicle dynamics to obtain a partially linearised input-output behaviour, while INDI improves robustness by formulating the inversion incrementally around the current measured or estimated motion of the vehicle. These methods have therefore been successfully applied to nonlinear flight-control problems and are well suited for cascaded hypersonic control architectures [5, 6]. However, inversion-based controllers are primarily tracking controllers. They are designed to make the vehicle follow commanded trajectories, but they do not inherently guarantee that safety-critical state constraints are respected during transient manoeuvres. In the context of scramjet-powered hypersonic flight, this limitation is important: even if the nominal controller achieves good tracking performance, it may still command a response that violates the admissible angle-of-attack magnitude or angle-of-attack-rate limits associated with inlet unstart.

This paper addresses this gap by augmenting an existing cascaded NDI/INDI flight-control architecture with a formal inlet-unstart protection mechanism. The proposed approach uses Control Barrier Functions (CBFs) to enforce safety constraints on the angle of attack and its rate of change. CBFs provide a mathematically rigorous framework for rendering a prescribed safe set forward invariant, thereby ensuring that system trajectories starting inside the admissible set remain inside it for all future time, provided the CBF constraint remains feasible [7]. When implemented through a Quadratic Programme (QP), the CBF acts as a minimally invasive safety filter: the nominal control command is passed through unchanged during normal operation and is modified only when necessary to prevent violation of the safety boundary. This makes the approach particularly suitable for integration with an existing flight-control architecture, since the baseline controller is preserved whenever the vehicle operates safely inside the admissible inlet-operating set.

CBFs have also been applied to safety-critical flight-control problems, most notably flight envelope protection and collision avoidance schemes [8–11]. This makes them a natural candidate for inlet-unstart protection, where violation of the admissible incidence conditions can have immediate and severe consequences. In this setting, the forward-invariance property provided by a CBF is particularly attractive, since it offers a formal mechanism for ensuring that the closed-loop response remains inside the prescribed inlet-operating set.

Several alternative methods exist for enforcing constraints in nonlinear control systems. Model Predictive Control (MPC) can handle state and input constraints explicitly by solving a constrained optimisation problem over a finite prediction horizon [12]. However, its computational cost can be significant, especially at the high update rates required for hypersonic attitude dynamics [13]. In addition, MPC performance depends strongly on the accuracy of the prediction model over the horizon, which is difficult to guarantee across the wide Mach-altitude envelope and uncertain aero-propulsive environment of a hypersonic vehicle. Barrier Lyapunov Functions (BLFs) offer another constraint-enforcement approach by embedding the constraint directly into a Lyapunov function that becomes singular at the boundary [14]. Although BLFs can provide a combined stability and safety certificate, their singular nature may induce increasingly aggressive corrective commands near the constraint boundary. For hypersonic vehicles, such aggressive commands are undesirable because they may drive actuators into saturation, potentially degrading controllability in precisely the regime where safety intervention is required. In contrast, the CBF-QP formulation enforces the constraint while explicitly minimising the deviation from the nominal command, thereby limiting unnecessary control activity.

The main contribution of this work is the formulation of a CBF-based inlet-unstart protection layer for a hypersonic NDI/INDI flight-control architecture. The protection problem is posed in terms of an admissible inlet-operating set defined by bounds on both the angle-of-attack magnitude and the angle-of-attack rate. The resulting safety filter modifies the nominal elevator-related command only when required to preserve this admissible set, while otherwise leaving the baseline controller unchanged. The proposed formulation therefore combines the tracking capability and robustness properties of the underlying NDI/INDI controller with a formal forward-invariance guarantee for inlet-unstart protection.

II. Theoretical Background

Consider an output space $\mathcal{Y} \subset \mathbb{R}^p$, state space $\mathcal{X} \subset \mathbb{R}^n$ and a control input space $\mathcal{U} \subset \mathbb{R}^m$, where $\mathcal{X}$ and $\mathcal{Y}$ are assumed to be path-connected and satisfy $\mathbf{0} \in \mathcal{X}$ and $\mathbf{0} \in \mathcal{Y}$. Consider a control-affine nonlinear system of the form

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + g(\boldsymbol{x})\,\boldsymbol{u}, \tag{1}$$

$$\boldsymbol{y} = z(\boldsymbol{x}), \tag{2}$$

where $\boldsymbol{x} \in \mathcal{X}$ is the state vector, $\boldsymbol{u} \in \mathcal{U}$ is the control input, and $\boldsymbol{y} \in \mathcal{Y}$ is the output vector, and $f : \mathbb{R}^n \to \mathbb{R}^n$, $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, and $z : \chi \to \mathbb{R}^n$ are locally Lipschitz continuous.

A. Nonlinear Dynamic Inversion

The foundational principle of feedback linearization, also referred to as Nonlinear Dynamic Inversion (NDI), is to linearize the input–output behavior of the control-affine plant dynamics established in Eq. (1) and Eq. (2) via exact state feedback and a coordinate transformation, bypassing the limitations of localized Jacobian linearization. This structural transformation maps the open-loop state dynamics into decoupled chains of integrators, thereby permitting the deployment of conventional linear control synthesis techniques [15].

By applying a suitable feedback-linearizing coordinate transformation to the underlying system dynamics in Eq. (1) and Eq. (2), the resulting input–output relation can be cast directly in terms of the pseudo-control vector $\boldsymbol{\nu}$ as

$$\boldsymbol{\nu} \;=\; \boldsymbol{f}_\nu(\boldsymbol{x}) + \boldsymbol{G}_\nu(\boldsymbol{x})\,\boldsymbol{u}, \tag{3}$$

where $\boldsymbol{f}_\nu(\boldsymbol{x})$ denotes the transformed drift vector field and $\boldsymbol{G}_\nu(\boldsymbol{x})$ the associated control effectiveness matrix. The pseudo-control vector $\boldsymbol{\nu} \in \mathbb{R}^p$ collects the higher-order derivatives of the controlled outputs up to their respective relative degrees, defined as [16]

$$\boldsymbol{\nu} \triangleq \begin{bmatrix} y_1^{(r_1)} & y_2^{(r_2)} & \cdots & y_p^{(r_p)} \end{bmatrix}^T, \tag{4}$$

where $y_i^{(r_i)}$ denotes the r_i-th time derivative of the i-th output, with r_i representing its corresponding relative degree relative to the input channel $\boldsymbol{u}$ in Eq. (1). To achieve a desired closed-loop tracking response, a target virtual control input $\boldsymbol{\tau}_{\text{des}} \in \mathbb{R}^p$ is introduced to represent the control-dependent component of the pseudo-control, such that $\boldsymbol{\tau}_{\text{des}} \triangleq \boldsymbol{\nu} - \boldsymbol{f}_\nu(\boldsymbol{x})$. This desired virtual input can be generated utilizing any standard linear control methodology. Assuming that the input matrix $\boldsymbol{G}_\nu(\boldsymbol{x})$ is square ($m = n$) and non-singular, the physical control input vector $\boldsymbol{u}$ driving the original system in Eq. (1) is directly recovered by inverting the input matrix mapping:

$$\boldsymbol{u} = \boldsymbol{G}_\nu^{-1}(\boldsymbol{x})\,\boldsymbol{\tau}_{\text{des}}. \tag{5}$$

B. Incrementalization of the Nonlinear Control Laws

In the presence of significant model unmodeled dynamics and aerodynamic uncertainties, purely model-based nonlinear control approaches can exhibit degraded robustness and performance. To mitigate this vulnerability, researchers have developed incremental formulations of these methods. This framework enables the partial substitution of explicit analytical model knowledge with real-time sensor measurements under relatively mild operational assumptions [17–19].

To derive this formulation, the nominal control-affine system in Eq. (1) is approximated by taking a first-order Taylor series expansion about a operating condition defined by the current state $\boldsymbol{x}_0$ and the corresponding control input $\boldsymbol{u}_0$. This linearization yields the following structural expression for the state derivative:

$$\begin{aligned}\dot{\boldsymbol{x}} \approx\ & f(\boldsymbol{x}_0) + g(\boldsymbol{x}_0)\,\boldsymbol{u}_0 + \frac{\partial}{\partial \boldsymbol{x}}\Big[f(\boldsymbol{x}) + g(\boldsymbol{x})\,\boldsymbol{u}\Big]\Big|_{\boldsymbol{x}_0,\boldsymbol{u}_0} (\boldsymbol{x} - \boldsymbol{x}_0) \\ & + \frac{\partial}{\partial \boldsymbol{u}}\Big[f(\boldsymbol{x}) + g(\boldsymbol{x})\,\boldsymbol{u}\Big]\Big|_{\boldsymbol{x}_0,\boldsymbol{u}_0} (\boldsymbol{u} - \boldsymbol{u}_0) + \text{H.O.T.},\end{aligned} \tag{6}$$

where H.O.T. denotes the higher-order terms, which are standardly neglected in flight control applications.

To introduce the foundational incremental concept simply, consider a system governed by Eq. (1) that exhibits a relative degree of one. The leading two terms on the right-hand side of Eq. (6) collectively represent the instantaneous state rate of change evaluated at the current time step, denoted as $\dot{\boldsymbol{x}}_0 \triangleq f(\boldsymbol{x}_0) + g(\boldsymbol{x}_0)\,\boldsymbol{u}_0$, which is assumed to be perfectly measurable via onboard inertial instrumentation. Furthermore, a strict time-scale separation is assumed between the fast actuator dynamics and the evolving state kinematics within a sufficiently small sampling interval Δt. This assumption implies that variations in the control channel dominate over state variations, such that $\|\boldsymbol{u} - \boldsymbol{u}_0\| \gg \|\boldsymbol{x} - \boldsymbol{x}_0\|$, thereby allowing the state-variation sensitivities to be safely neglected. Assuming complete, instantaneous state observability ($\boldsymbol{y} = \boldsymbol{x}$) and defining the incremental control input as $\Delta\boldsymbol{u} \triangleq \boldsymbol{u} - \boldsymbol{u}_0$, Eq. (6) reduces to the simplified incremental plant dynamics:

$$\dot{\boldsymbol{x}} \approx \dot{\boldsymbol{x}}_0 + g(\boldsymbol{x}_0)\,\Delta\boldsymbol{u}. \tag{7}$$

By matching the left-hand side to a target pseudo-control input $\boldsymbol{\nu}$ using the framework discussed in Section II.C, we obtain the Incremental Nonlinear Dynamic Inversion (INDI) control law:

$$\Delta\boldsymbol{u} = g^{-1}(\boldsymbol{x}_0)\,\big(\boldsymbol{\nu} - \dot{\boldsymbol{x}}_0\big). \tag{8}$$

Because the incremental law in Eq. (8) specifies only the localized variation in command, the total physical control command applied to the primary plant in Eq. (1) is updated sequentially by adding this increment to the current baseline control state:

$$\boldsymbol{u} = \boldsymbol{u}_0 + \Delta\boldsymbol{u}. \tag{9}$$

C. Control Barrier Functions

To define safety for system (1), we consider a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$ and a set C defined as the zero-superlevel set of h, yielding

$$C \triangleq \big\{\boldsymbol{x} \in \mathbb{R}^n : h(\boldsymbol{x}) \geq 0\big\}. \tag{10}$$

Definition II.1 (Forward Invariance and safety, [20])**.** *The set C is said to be* forward invariant *with respect to system* (1) *if, for every initial condition $\boldsymbol{x}(0) \in C$, it follows that $\mathbf{x}(t) \in S$ for all $t \in I(\mathbf{x}_0) = [0, \tau_{\max} = \infty)$. The system* (1) *is safe with respect to S if S is forward invariant [7].*

We introduce the concept of a Control Barrier Function (CBF), which will be used to ensure that the system can be rendered safe with respect to S via the control input $\mathbf{u}$ [7].

Definition II.2 (Control Barrier Function [7])**.** *Let $C \subset \mathbb{R}^n$ be a safe set defined by a continuously differentiable function h. Then h is a CBF for system* (1) *on C if there exists a class $\mathcal{K}$ function $\alpha : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ such that, for all $\boldsymbol{x} \in C$,*

$$\sup_{\boldsymbol{u} \in \mathcal{U}} \Big[\underbrace{L_f h(\boldsymbol{x})}_{\nabla h \cdot f} + \underbrace{L_g h(\boldsymbol{x})}_{\nabla h \cdot g} \; \boldsymbol{u} \Big] \;\geq\; -\alpha\big(h(\boldsymbol{x})\big), \tag{11}$$

where $L_f h(\boldsymbol{x}) = \nabla h(\boldsymbol{x}) \cdot f(\boldsymbol{x})$ and $L_g h(\boldsymbol{x}) = \nabla h(\boldsymbol{x}) \cdot g(\boldsymbol{x})$ denote the Lie derivatives of h along f and g, respectively.

Theorem II.1 (Safety Guarantee [7])**.** *If h is a CBF for system* (1) *on C, then any locally Lipschitz control law $\boldsymbol{u} = \kappa(\boldsymbol{x})$ satisfying condition* (11) *renders C forward invariant.*

For safety constraints where the CBF candidate $h(\boldsymbol{x})$ possesses a relative degree greater than one with respect to the system dynamics (1), standard CBF formulations cannot directly enforce safety. To handle such constraints on higher-order state variables, the classical formulation is generalized to High-Order Control Barrier Functions (HOCBFs) [21].

Definition II.3 (High-Order Control Barrier Function [21])**.** *Let $h : \mathbb{R}^n \to \mathbb{R}$ be a d-times continuously differentiable function. Define a sequence of functions $\psi_i : \mathbb{R}^n \to \mathbb{R}$ recursively as follows:*

$$\begin{aligned}
\psi_0(\boldsymbol{x}) \;&\triangleq\; h(\boldsymbol{x}), \\
\psi_1(\boldsymbol{x}) \;&\triangleq\; \dot{\psi}_0(\boldsymbol{x}) + \alpha_1\big(\psi_0(\boldsymbol{x})\big), \\
&\;\vdots \\
\psi_d(\boldsymbol{x}) \;&\triangleq\; \dot{\psi}_{d-1}(\boldsymbol{x}) + \alpha_d\big(\psi_{d-1}(\boldsymbol{x})\big),
\end{aligned} \tag{12}$$

where $\alpha_1(\cdot), \ldots, \alpha_d(\cdot)$ denote class $\mathcal{K}$ functions. Then $h(\boldsymbol{x})$ is a High-Order Control Barrier Function *of relative degree d for system* (1) *if there exist class $\mathcal{K}$ functions $\alpha_1, \ldots, \alpha_d$ such that, for all $\boldsymbol{x} \in C$,*

$$\sup_{\boldsymbol{u} \in \mathcal{U}} \Big[L_f^d h(\boldsymbol{x}) + L_g L_f^{d-1} h(\boldsymbol{x})\, \boldsymbol{u} + O\big(h(\boldsymbol{x})\big) + \alpha_d\big(\psi_{d-1}(\boldsymbol{x})\big) \Big] \;\geq\; 0, \tag{13}$$

where $O\big(h(\boldsymbol{x})\big) \triangleq \sum_{i=1}^{d-1} L_f^i\big(\alpha_{d-i}(\psi_{d-i-1}(\boldsymbol{x}))\big)$ collects the remaining Lie derivatives along $f(\boldsymbol{x})$ resulting from the recursive expansion up to degree $d-1$.

Analogous to Theorem II.1, the existence of an HOCBF guarantees that the control input $\boldsymbol{u}$ can render the high-order safe set forward invariant, bypassing the relative degree limitation by structurally propagating the constraint through successive state derivatives.

III. The Inlet Unstart Protection Problem

The scramjet inlet operates by compressing incoming supersonic air through a system of oblique shocks before delivering it to the combustion chamber. As illustrated in Fig. 1, under nominal flight conditions the forebody compression ramp generates an oblique shock that impinges precisely on the cowl lip, directing the compressed flow into the isolator. A stable bifurcated shock train is then maintained within the isolator, providing the mass flow and total pressure recovery required for sustained combustion [2]. This shock train is sensitive to the local angle of incidence at the inlet face. A sufficiently large excursion in angle of attack shifts the forebody shock impingement point beyond the cowl lip, causing the shock train to be expelled upstream from the isolator in a process known as *inlet unstart*.

The consequences of unstart are severe and rapid, as illustrated in Fig. 2. The upstream propagation of the shock train out of the isolator creates a standing normal shock ahead of the inlet entrance, effectively blocking supersonic air

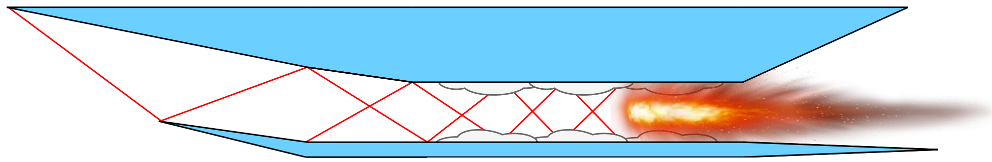

Fig. 1 Cross-section of a scramjet chin inlet under nominal started conditions.

from entering the engine [2]. This causes an abrupt and simultaneous loss of thrust, increase in drag, decrease in lift, and an impulsive change in pitching moment. For a vehicle with two or more engines, an asymmetric unstart event introduces large yawing and rolling moments that the flight control system must counteract [22]. At hypersonic flight conditions, where dynamic pressure is high and control authority is limited by structural and thermal constraints, these disturbances can exceed the stabilising capability of the control system and result in vehicle divergence.

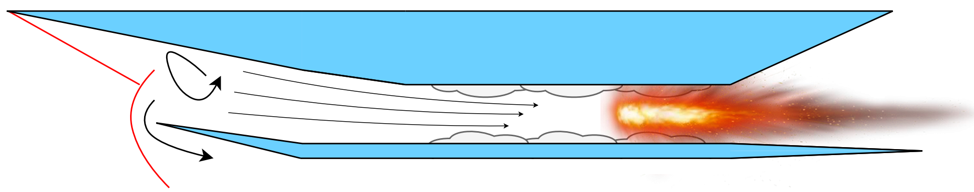

Fig. 2 Schematic illustration of a scramjet chin inlet during inlet unstart.

The angle of attack at which unstart occurs under steady, slowly-varying flight conditions defines the *static unstart boundary**. This boundary is not fixed but decreases with increasing Mach number, as the forebody shock becomes more strongly swept and the tolerance of the inlet contraction ratio narrows. Experimental studies have reported unstart onset at angles of attack as low as 6° to 8° at Mach 5 for chin-inlet configurations [23], which provides a reasonable reference boundary for the GHAME vehicle as it also employs a chin-inlet design.

The magnitude of the angle of attack alone does not fully determine whether unstart occurs. How quickly the angle of attack changes matters equally, as the shock train within the isolator requires a finite time to reposition in response to a change in incidence. At low pitch rates this adaptation is quasi-steady and the static boundary remains a reasonable approximation. At higher pitch rates the shock train cannot follow the changing incidence fast enough, causing unstart to occur at a lower angle of attack than the static boundary would predict, with onset becoming earlier and more abrupt as pitch rate increases [24]. Computational simulations over an angle-of-attack range of 0° to 10° at pitch rates of 10 deg/s and 100 deg/s have confirmed this dependence directly [24].

Once unstart occurs, recovery is not straightforward. The inlet exhibits a hysteresis phenomenon in which the angle of attack must be reduced significantly below the original unstart threshold before the shock train can re-establish itself [1]. At high Mach numbers this recovery may require a reduction in angle of attack that is infeasible within the available flight envelope, making prevention the only reliable strategy. To prevent inlet unstart, the angle-of-attack rate must therefore be kept below a flight-condition-dependent bound. Based on the literature, this bound is on the order of 5 deg/s for hypersonic cruise conditions, with stricter limits required at higher Mach numbers where the shock system is more sensitive to incidence changes [1, 24].

The objective of this paper is to design a control augmentation mechanism for the existing NDI/INDI cascade of [25] that protects the vehicle from inlet-unstart operating conditions while preserving attitude-command tracking as closely as possible. The baseline cascade receives attitude commands from a higher-level flight-path controller; however, aggressive command tracking may induce large or rapid variations in angle of attack and thereby drive the vehicle toward critical flow conditions of the scramjet inlet.

Accordingly, the augmented closed-loop system shall satisfy

$$\alpha_{\min} \leq \alpha(t) \leq \alpha_{\max}, \qquad \dot{\alpha}_{\min} \leq \dot{\alpha}(t) \leq \dot{\alpha}_{\max}, \qquad \forall t \geq t_0, \tag{14}$$

for all initially admissible conditions. The bounds on α define the admissible incidence envelope, while the bounds on

*The term static refers to conditions where the angle of attack changes slowly enough that the shock train can fully adapt at each instant, so the inlet behaviour is determined by the instantaneous value of α alone.

$\dot{\alpha}$ limit rapid changes in the flow conditions relevant to inlet unstart. The resulting control mechanism should intervene only when required to maintain these constraints, and otherwise retain the nominal behavior of the NDI/INDI cascade.

IV. Nonlinear Control Architecture with Inlet Unstart Protection

This section presents the nonlinear flight control architecture integrated for the GHAME vehicle and introduces the CBF-based inlet-unstart protection layer employed in the present work. The baseline control system follows the cascaded nonlinear dynamic inversion (NDI) and incremental nonlinear dynamic inversion (INDI) architecture developed in [25] and is not redesigned in the present work. In its complete form, this architecture consists of four cascaded loops, namely position, flight-path, attitude, and angular-rate control. The position and attitude loops employ NDI, exploiting the exact kinematic relations available at these levels, whereas the flight-path and angular-rate loops use INDI to compensate for aerodynamic uncertainty acting on the translational and rotational dynamics. The complete cascaded structure, including the safety filter, is shown in Fig. 3.

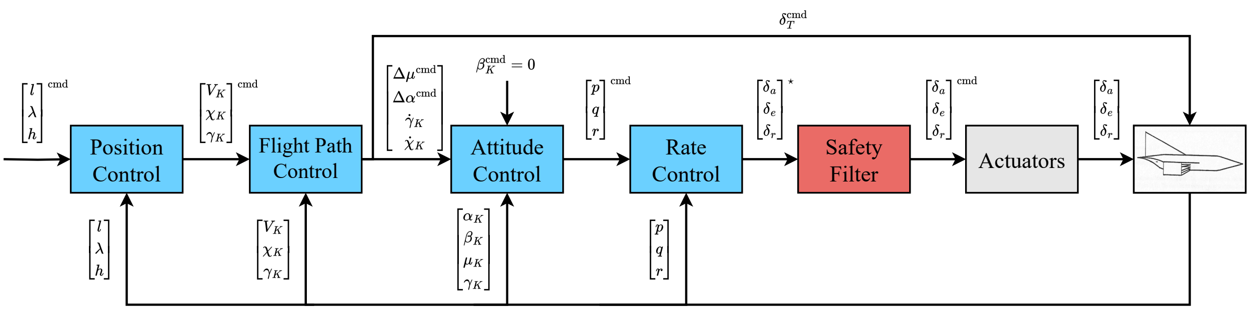

Fig. 3 Four-loop cascaded NDI/INDI flight control architecture from [25], extended with a CBF-based safety filter.

The extension introduced in this work addresses the inlet-unstart protection problem, which is not explicitly considered in the baseline control architecture proposed in [25]. As discussed in Section III, inlet unstart represents a critical failure mode for air-breathing hypersonic vehicles and may occur when excessive incidence magnitudes or rapid incidence variations destabilise the internal shock system. Although the baseline controller provides nonlinear flight control over the considered operating regime, it does not impose formal constraints on the angle-of-attack conditions relevant to inlet operation. The present work therefore introduces a CBF-based protection layer that enforces prescribed bounds on both the angle of attack α and its rate of change $\dot{\alpha}$. Within the validity range of the employed model, this formulation renders the admissible inlet-operating set forward invariant and thereby provides guaranteed protection against inlet unstart.

A. Nonlinear Attitude Control Laws

The attitude-control subsystem is formulated as a cascaded aerodynamic-angle and angular-rate controller. The design relies on a time-scale separation assumption, according to which the inner angular-rate dynamics are sufficiently faster than the outer aerodynamic-angle dynamics. The outer loop converts commanded aerodynamic-angle motion into commanded body rates, while the inner loop computes the moment increments and corresponding control-surface commands required to track these rates. This cascaded formulation is particularly relevant for the subsequent inlet-unstart protection layer, since both the angle of attack and its rate of change are determined by the attitude response and by the elevator channel generated through the inner-loop control allocation. Let the aerodynamic-angle vector and its commanded value be defined as

$$\boldsymbol{\eta} = \begin{bmatrix} \mu & \alpha & \beta \end{bmatrix}^\top, \qquad \boldsymbol{\eta}_{\mathrm{cmd}} = \begin{bmatrix} \mu_{\mathrm{cmd}} & \alpha_{\mathrm{cmd}} & \beta_{\mathrm{cmd}} \end{bmatrix}^\top, \tag{15}$$

where μ, α, and β denote bank angle, angle of attack, and sideslip angle, respectively. The aerodynamic-angle tracking error is defined as

$$\boldsymbol{e}_\eta = \boldsymbol{\eta}_{\mathrm{cmd}} - \boldsymbol{\eta}. \tag{16}$$

The outer-loop controller computes the commanded aerodynamic-angle-rate vector by proportional feedback,

$$\dot{\boldsymbol{\eta}}_{\text{cmd}} = K_\eta \boldsymbol{e}_\eta, \qquad K_\eta = \begin{bmatrix} K_\mu & 0 & 0 \\ 0 & K_\alpha & 0 \\ 0 & 0 & K_\beta \end{bmatrix}. \tag{17}$$

Here, K_η is a diagonal gain matrix that assigns independent closed-loop gains to the bank-angle, angle-of-attack, and sideslip-angle channels.

The aerodynamic-angle kinematics are given by

$$\begin{bmatrix} \dot{\mu} \\ \dot{\alpha} \\ \dot{\beta} \end{bmatrix} = \left(\frac{1}{mV}\right) \underbrace{\begin{bmatrix} \cos\gamma\cos\mu\tan\beta & \tan\gamma\sin\mu + \tan\beta & \tan\gamma\cos\mu\cos\beta \\ \cos\gamma\cos\mu/\cos\beta & -1/\cos\beta & 0 \\ \cos\gamma\sin\mu & 0 & \cos\beta\cos\gamma \end{bmatrix}}_{T_F(\boldsymbol{x})} \begin{bmatrix} -W \\ L \\ Y \end{bmatrix} + \underbrace{\begin{bmatrix} \cos\alpha/\cos\beta & 0 & \sin\alpha/\cos\beta \\ -\cos\alpha\tan\beta & 1 & -\sin\alpha\tan\beta \\ \sin\alpha & 0 & -\cos\alpha \end{bmatrix}}_{T_\omega(\alpha,\beta)} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \tag{18}$$

In compact notation, Eq. (18) can be written as

$$\dot{\boldsymbol{\eta}} = \frac{1}{mV} T_F(\boldsymbol{x}) \begin{bmatrix} -W \\ L \\ Y \end{bmatrix} + T_\omega(\alpha,\beta)\boldsymbol{\omega}, \qquad \boldsymbol{\omega} = \begin{bmatrix} p & q & r \end{bmatrix}^\top. \tag{19}$$

Here, m denotes the vehicle mass, V the airspeed, $W = mg$ the weight, g the gravitational acceleration, L the aerodynamic lift force, Y the aerodynamic side force, and γ the flight-path angle. The matrix $T_F(\boldsymbol{x})$ captures the contribution of the force balance to the aerodynamic-angle rates, whereas $T_\omega(\alpha,\beta)$ maps the body rates into the corresponding aerodynamic-angle rates. The commanded body-rate vector is obtained by imposing $\dot{\boldsymbol{\eta}} = \dot{\boldsymbol{\eta}}_{\text{cmd}}$ in Eq. (19) and solving for $\boldsymbol{\omega}$. Assuming that $T_\omega(\alpha,\beta)$ is nonsingular at the considered operating condition, this gives

$$\boldsymbol{\omega}_{\text{cmd}} = T_\omega^{-1}(\alpha,\beta) \left[\dot{\boldsymbol{\eta}}_{\text{cmd}} - \frac{1}{mV} T_F(\boldsymbol{x}) \begin{bmatrix} -W \\ L \\ Y \end{bmatrix} \right], \tag{20}$$

with

$$\boldsymbol{\omega}_{\text{cmd}} = \begin{bmatrix} p_{\text{cmd}} & q_{\text{cmd}} & r_{\text{cmd}} \end{bmatrix}^\top. \tag{21}$$

The inner angular-rate loop tracks $\boldsymbol{\omega}_{\text{cmd}}$. The rate-tracking error is defined as

$$\boldsymbol{e}_\omega = \boldsymbol{\omega}_{\text{cmd}} - \boldsymbol{\omega}. \tag{22}$$

A proportional rate controller computes the commanded virtual angular acceleration,

$$\boldsymbol{\nu}_{\text{cmd}} = K_\omega \boldsymbol{e}_\omega, \qquad K_\omega = \begin{bmatrix} K_p & 0 & 0 \\ 0 & K_q & 0 \\ 0 & 0 & K_r \end{bmatrix}, \tag{23}$$

where

$$\boldsymbol{\nu}_{\text{cmd}} = \begin{bmatrix} \dot{p}_{\text{cmd}} & \dot{q}_{\text{cmd}} & \dot{r}_{\text{cmd}} \end{bmatrix}^\top. \tag{24}$$

To reduce the dependence on an exact rotational aerodynamic model, the rate-loop inversion is implemented in incremental form. The currently acting virtual angular acceleration is estimated from the available body-rate signals, coming from a gyroscope, using numerical differentiation and low-pass filtering. Denoting this estimate by

$$\boldsymbol{\nu}_0 = \begin{bmatrix} \dot{p}_0 & \dot{q}_0 & \dot{r}_0 \end{bmatrix}^\top, \tag{25}$$

the commanded virtual-control increment is defined as

$$\Delta \boldsymbol{\nu}_{\text{cmd}} = \boldsymbol{\nu}_{\text{cmd}} - \boldsymbol{\nu}_0. \tag{26}$$

The incremental formulation therefore commands only the additional angular acceleration required relative to the estimated current rotational response. The required incremental moment vector follows from the rigid-body rotational dynamics. With the inertia matrix I, the commanded moment increment is written as

$$\Delta \boldsymbol{Q}_{\text{cmd}} = \begin{bmatrix} \Delta L_{\text{cmd}} \\ \Delta M_{\text{cmd}} \\ \Delta N_{\text{cmd}} \end{bmatrix} = I \Delta \boldsymbol{\nu}_{\text{cmd}} + \boldsymbol{\omega} \times I \boldsymbol{\omega}. \tag{27}$$

The incremental moment command is then mapped to incremental control-surface commands through a control mixing logic. The vehicle uses two elevons and one rudder as aerodynamic control surfaces. The left and right elevon deflections are denoted by δ_{vl} and δ_{vr}, respectively, while the rudder deflection is denoted by δ_r. The symmetric and antisymmetric elevon combinations define the elevator and aileron channels,

$$\delta_e = \frac{\delta_{vl} + \delta_{vr}}{2}, \qquad \delta_a = \frac{\delta_{vl} - \delta_{vr}}{2}. \tag{28}$$

Let the incremental control-surface vector be

$$\Delta \boldsymbol{\delta}_{\text{cmd}} = \begin{bmatrix} \Delta\delta_a & \Delta\delta_e & \Delta\delta_r \end{bmatrix}^\top. \tag{29}$$

With the local control-effectiveness matrix B, the incremental control command is computed as

$$\Delta \boldsymbol{\delta}_{\text{cmd}} = B^{+} \Delta \boldsymbol{Q}_{\text{cmd}} = B^T \left(B B^T \right)^{-1} \Delta \boldsymbol{Q}_{\text{cmd}}. \tag{30}$$

The final control command is obtained by adding the incremental command to the estimated current actuator state,

$$\boldsymbol{\delta}_{\text{cmd}} = \boldsymbol{\delta}_0 + \Delta \boldsymbol{\delta}_{\text{cmd}}. \tag{31}$$

Here, $\boldsymbol{\delta}_0$ denotes the internally estimated actuator state about which the incremental command is applied. The actuator dynamics are represented by a second-order transfer function with position and rate limits,

$$\frac{\delta(s)}{\delta_{\text{cmd}}(s)} = \frac{\omega_{\text{act}}^2}{s^2 + 2\zeta_{\text{act}}\omega_{\text{act}} s + \omega_{\text{act}}^2}, \tag{32}$$

where ω_{act} is the actuator natural frequency and ζ_{act} is the damping ratio. The actuator model is used to propagate the internal estimate $\boldsymbol{\delta}_0$, which closes the incremental control loop around the estimated actuator state rather than around the commanded surface deflection alone.

B. CBF-Based Safety Filter for Inlet Unstart Protection

For the inlet-unstart protection problem, the relevant longitudinal relation is obtained from Eq. (18) by assuming coordinated flight,

$$\beta = 0, \qquad \mu = 0, \qquad p = r = 0. \tag{33}$$

Extracting the $\dot{\alpha}$-row gives

$$\dot{\alpha} = q - \frac{L + W\cos\gamma}{mV}. \tag{34}$$

Thus, the angle-of-attack rate is directly influenced by the pitch-rate response generated by the inner loop and by the aerodynamic force balance normal to the velocity vector.To expose the dependence on the elevator channel, the lift force is written as

$$L = \bar{q} S \left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_{\delta_e}}\delta_e \right), \tag{35}$$

where $\bar{q} = \frac{1}{2}\rho V^2$ is the dynamic pressure, S the reference surface area, C_{L_0} the zero-incidence lift coefficient, C_{L_α} the lift-curve slope, and $C_{L_{\delta_e}}$ the elevator lift effectiveness. Substituting Eq. (35) into Eq. (34) yields

$$\dot{\alpha} = q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_{\delta_e}}\delta_e\right) - \frac{g\cos\gamma}{V}. \tag{36}$$

Equation (36) links the elevator channel to the angle-of-attack-rate response, while α itself defines the instantaneous incidence magnitude. These two quantities are therefore used in the subsequent CBF formulation to impose inlet-unstart protection constraints on both α and $\dot{\alpha}$.

From Eq. (36), the angle-of-attack rate estimate used by the filter is obtained by neglecting the gravity term $g\cos\gamma/V$, which is small relative to the aerodynamic terms at hypersonic cruise conditions, giving

$$\dot{\alpha}_{\text{est}} \;\approx\; q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_{\delta_e}}\delta_e\right). \tag{37}$$

The dependence of $\dot{\alpha}_{\text{est}}$ on δ_e is affine, with control effectiveness

$$\frac{\partial\dot{\alpha}_{\text{est}}}{\partial\delta_e} \;=\; -\frac{\bar{q}S\,C_{L_{\delta_e}}}{mV}. \tag{38}$$

Define the upper barrier function in terms of the estimated angle-of-attack rate, where the state-dependent terms q, α, and $\bar{q}$ are treated as measured quantities at the current timestep:

$$h(\delta_e) \;\triangleq\; \dot{\alpha}_{\max} - \dot{\alpha}_{\text{est}}(\delta_e), \tag{39}$$

so that the safe set of Eq. (10) becomes

$$C \;=\; \left\{\delta_e \in \mathbb{R} : \dot{\alpha}_{\text{est}}(\delta_e) \le \dot{\alpha}_{\max}\right\}. \tag{40}$$

Since $\dot{\alpha}_{\text{est}}$ is affine in δ_e, the relative degree of h with respect to δ_e is unity. Under the assumption that the flight condition varies slowly relative to the elevator dynamics, the state-dependent terms in Eq. (37) are treated as constant over one sample interval. Taking the time derivative of Eq. (39) and substituting Eq. (38) then yields

$$\dot{h} \;=\; -\frac{\partial\dot{\alpha}_{\text{est}}}{\partial\delta_e}\,\dot{\delta}_e \;=\; \frac{\bar{q}S\,C_{L_{\delta_e}}}{mV}\,\dot{\delta}_e. \tag{41}$$

Substituting Eq. (41) into the CBF condition (11) with linear gain $k_1 > 0$ gives

$$\frac{\bar{q}S\,C_{L_{\delta_e}}}{mV}\,\dot{\delta}_e + k_1\,h(\delta_e) \;\ge\; 0, \tag{42}$$

which, after rearrangement, yields the upper rate constraint on the admissible elevator rate:

$$\boxed{\dot{\delta}_e \;\ge\; -k_1\,\frac{mV}{\bar{q}S\,C_{L_{\delta_e}}}\left(\dot{\alpha}_{\max} - \dot{\alpha}_{\text{est}}\right).} \tag{43}$$

The lower bound $\dot{\alpha} \ge \dot{\alpha}_{\min}$ is treated symmetrically by defining $h_-(\delta_e) = \dot{\alpha}_{\text{est}} - \dot{\alpha}_{\min}$, which yields the floor constraint

$$\boxed{\dot{\delta}_e \;\le\; k_1\,\frac{mV}{\bar{q}S\,C_{L_{\delta_e}}}\left(\dot{\alpha}_{\text{est}} - \dot{\alpha}_{\min}\right).} \tag{44}$$

Remark IV.1. *The permissible rate of change of δ_e is proportional to the remaining margin $\dot{\alpha}_{\max} - \dot{\alpha}_{\text{est}}$: the constraint is inactive when $\dot{\alpha}_{\text{est}}$ is well within the bound and becomes increasingly restrictive as $\dot{\alpha}_{\text{est}} \to \dot{\alpha}_{\max}$, asymptotically preventing boundary crossing.*

Beyond constraining $\dot{\alpha}$, the angle of attack itself must remain within the aerodynamic envelope defined by Eq. (14). Since $\dot{\alpha}_{\text{est}}$ in Eq. (37) depends affinely on δ_e, the time derivative of any function of α will also contain δ_e explicitly. The relative degree of the α barrier with respect to δ_e is therefore unity and a standard CBF applies.

Define the upper barrier function on α:

$$h_\alpha(\alpha) \triangleq \alpha_{\max} - \alpha, \tag{45}$$

with associated safe set

$$\mathcal{C}_\alpha = \left\{ \alpha \in \mathbb{R} : \alpha \le \alpha_{\max} \right\}. \tag{46}$$

Taking the time derivative of Eq. (45)and substituting Eq. (37), under the same slowly-varying flight condition assumption as before, yields

$$\dot{h}_\alpha = -\dot{\alpha}_{\text{est}} = -q + \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_{\delta_e}}\delta_e\right). \tag{47}$$

Applying the CBF condition (11) with linear gain $k_2 > 0$ gives

$$\dot{h}_\alpha + k_2\, h_\alpha \ge 0, \tag{48}$$

which, substituting Eqs. (45) and (47) and solving for δ_e, yields the upper envelope constraint:

$$\boxed{\delta_e \le \frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left[q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha\right) - k_2(\alpha_{\max} - \alpha)\right].} \tag{49}$$

The lower bound $\alpha \ge \alpha_{\min}$ is treated symmetrically by defining $h_{\alpha,-}(\alpha) = \alpha - \alpha_{\min}$, which yields the floor constraint

$$\boxed{\delta_e \ge \frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left[q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha\right) + k_2(\alpha - \alpha_{\min})\right].} \tag{50}$$

Remark IV.2. *In contrast to Eqs.* (43) *and* (44)*, which constrain the rate of elevator motion, Eqs.* (49) *and* (50) *impose a direct bound on the elevator deflection magnitude at each instant. The gain k_2 determines how early the constraint activates: larger values begin restricting δ_e further from the boundary, smaller values permit a closer approach before intervention [26].*

The rate constraints (43) and (44) are expressed in terms of δ_e by applying a first-order backward difference approximation,

$$\dot{\delta}_e \approx \frac{\delta_e - \delta_{e,k-1}^{\text{safe}}}{\Delta t}, \tag{51}$$

where $\delta_{e,k-1}^{\text{safe}}$ is the safe elevator command from the previous timestep and Δt is the sample period. All five constraints are then linear in δ_e, and the CBF constraints (43), (44), (49), and (50) are imposed simultaneously via a Quadratic Programme (QP) that minimally modifies the nominal elevator command δ_e^{cmd} produced by the angular rate loop:

$$\delta_e^{\text{safe}} = \underset{\delta_e \in \mathbb{R}}{\arg\min}\ \tfrac{1}{2}\left\|\delta_e - \delta_e^{\text{cmd}}\right\|_2^2 \tag{52}$$

$$\begin{aligned}
\text{subject to}\quad \delta_e &\ge \delta_{e,k-1}^{\text{safe}} - k_1\,\frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left(\dot{\alpha}_{\max} - \dot{\alpha}_{\text{est}}\right)\Delta t, && (53)\\
\delta_e &\le \delta_{e,k-1}^{\text{safe}} + k_1\,\frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left(\dot{\alpha}_{\text{est}} - \dot{\alpha}_{\min}\right)\Delta t, && (54)\\
\delta_e &\le \frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left[q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha\right) - k_2(\alpha_{\max} - \alpha)\right], && (55)\\
\delta_e &\ge \frac{mV}{\bar{q}S\, C_{L_{\delta_e}}}\left[q - \frac{\bar{q}S}{mV}\left(C_{L_0} + C_{L_\alpha}\alpha\right) + k_2(\alpha - \alpha_{\min})\right], && (56)\\
\delta_{e,\min} &\le \delta_e \le \delta_{e,\max}. && (57)
\end{aligned}$$

Remark IV.3. *In rare cases the feasible interval may become empty, for example when α is near $\alpha_{\max}$ and the α upper constraint demands a small or negative δ_e, while simultaneously the $\dot{\alpha}$ lower constraint prevents δ_e from decreasing further. In this situation the α constraints take priority over the $\dot{\alpha}$ constraints, since exceeding the aerodynamic envelope is more consequential than a transient rate violation.*

Remark IV.4. *The control effectiveness $C_{L_{\delta_e}}$ in all expressions varies with Mach number and dynamic pressure and is evaluated at the current flight condition from the aerodynamic dataset. This scheduling requires no additional sensors beyond those already available in the cascade.*

V. Outlook

The theoretical framework and mathematical derivation of the CBF-based safety filter have been presented in the preceding sections. The remaining work required to complete this paper concerns the implementation of the filter within the existing simulation environment and the collection and analysis of the resulting simulation results.

The safety filter derived in Section IV.B will be implemented in the MATLAB and Simulink simulation framework developed in [25]. The filter is to be inserted at the output of the angular rate loop, immediately before the actuator block, as described in Section IV.A. Once implemented, simulations will be conducted to evaluate the performance of the safety filter across representative manoeuvres that would, without the filter, produce angle-of-attack magnitudes or rates exceeding the prescribed bounds. The results are expected to demonstrate that the constraints set by (14) are enforced at all times, that the filter intervenes minimally and only when a constraint boundary is approached.

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